Quantum Algorithms for Optimization Problems

Evaluating QAOA and Recursive QAOA for Max-Cut: Depth, Approximation Ratio, and Solution Distribution

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Introduction

Combinatorial optimization problems are central to science and industry, with applications ranging from network design to machine learning. Among these, the Max-Cut problem (partitioning graph vertices into two sets to maximize crossing edges) is a well-known NP-hard problem and a standard benchmark for near-term quantum algorithms. Classical approaches can efficiently solve small instances but face scalability challenges for larger or complex graphs. The **Quantum Approximate Optimization Algorithm (QAOA)** provides a hybrid quantum-classical framework designed for noisy intermediate-scale quantum devices. However, standard local QAOA often suffers from parameter optimization overhead and limited performance at low circuit depths. To address these limitations, this work incorporates **Recursive QAOA (RQAOA)**, a recursive problem-reduction strategy, and presents a direct comparison between QAOA and RQAOA based on solution quality and computational trade-offs.

Problem Formulation & Objectives

1. Problem Statement

Given a graph $G(V, E)$ the Max-Cut problem aims to partition the vertices into two disjoint sets such that the number of edges between the sets is maximized. This problem is NP-hard and serves as a standard benchmark for evaluating optimization algorithms.

- **Mathematical Formulation of Max-Cut**

This can be expressed mathematically by assigning a binary variable $\boldsymbol{z_i} \in \{-\mathbf{1}, +\mathbf{1}\}$ to each vertex, where the two values represent the two partitions. The cut value is then given by:

$$\boldsymbol{C(z)} = \sum_{(i,j)\in E} \frac{\mathbf{1} - \boldsymbol{z_i z_j}}{\mathbf{2}}, \quad \boldsymbol{z_i} \in \{-\mathbf{1}, +\mathbf{1}\}$$

The term $\frac{1-z_i z_j}{2}$ evaluates to 1 if $z_i \neq z_j$ (edge is cut) and equals to 0 if $\boldsymbol{z_i = z_j}$ (edge is not cut). Maximizing $C(z)$ yields the maximum cut.

- **Hamiltonian Formulation**

The QAOA cost Hamiltonian $\boldsymbol{H_C}$ encodes the Max-Cut objective, while the mixer Hamiltonian $\boldsymbol{H_M}$ drives transitions between computational basis states:

$$\boldsymbol{H_C} = \sum_{(i,j)\in E} \frac{\mathbf{1} - \boldsymbol{Z_i Z_j}}{\mathbf{2}}$$

$$\boldsymbol{H_M} = \sum_{i} \boldsymbol{X_i}$$

Here, $\boldsymbol{Z_i}$ and $\boldsymbol{X_i}$ are Pauli operators acting on qubit $\boldsymbol{i}$. The cost Hamiltonian is diagonal in the computational basis and its expectation value gives the cut value to be maximized. The mixer Hamiltonian induces bit flips to explore the solution space.

2. Objective

The objective of this work focus on

$$\boldsymbol{max}_{(\gamma,\beta)} \langle \Psi(\gamma, \beta)|\boldsymbol{H_C}|\Psi(\gamma, \beta)\rangle$$

The classical optimizer updates γ and β to maximize the expectation value.

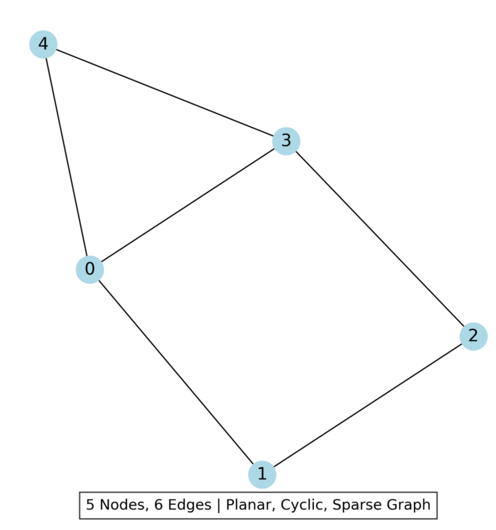
Problem Setup & Evaluation Metrics

This section describes the test graph, the mathematical formulation of Max-Cut, and the three metrics used to compare QAOA and RQAOA.

1. Graph Instance

We evaluate both algorithms on a single, well-defined graph to enable controlled comparison.

- **Nodes:** 05
- **Edges:** 06
- **Type:** Simple, undirected, connected
- **Structure:** Planar, cyclic



2. Performance Metrics

To enable fair comparison between QAOA, RQAOA, and classical baselines, we adopt the following metrics.

- **Approximation Ratio**

$$AR = C_{\text{found}} / C_{\text{optimal}}$$

- **Success Probability**

  Probability of measuring the optimal cut

- **Runtime**

  Total simulation + classical optimization time

Methodology: QAOA & RQAOA Framework

- **Quantum Approximate Optimization Algorithm**

QAOA is a hybrid quantum-classical algorithm that prepares a parameterized quantum state by alternating between cost and mixer Hamiltonians.

$$|\boldsymbol{\psi(\gamma,\beta)}\rangle = \prod_{k=1}^{p} \boldsymbol{e^{-i\beta_k H_M} e^{-i\gamma_k H_C}} |+\rangle^{\otimes n}$$

- **Recursive QAOA (RQAOA)**

RQAOA extends QAOA by iteratively reducing the problem size using correlation information.

Procedure:

- Run QAOA on current graph
- Compute correlations:

$$M_{ij} = \langle Z_i Z_j \rangle$$

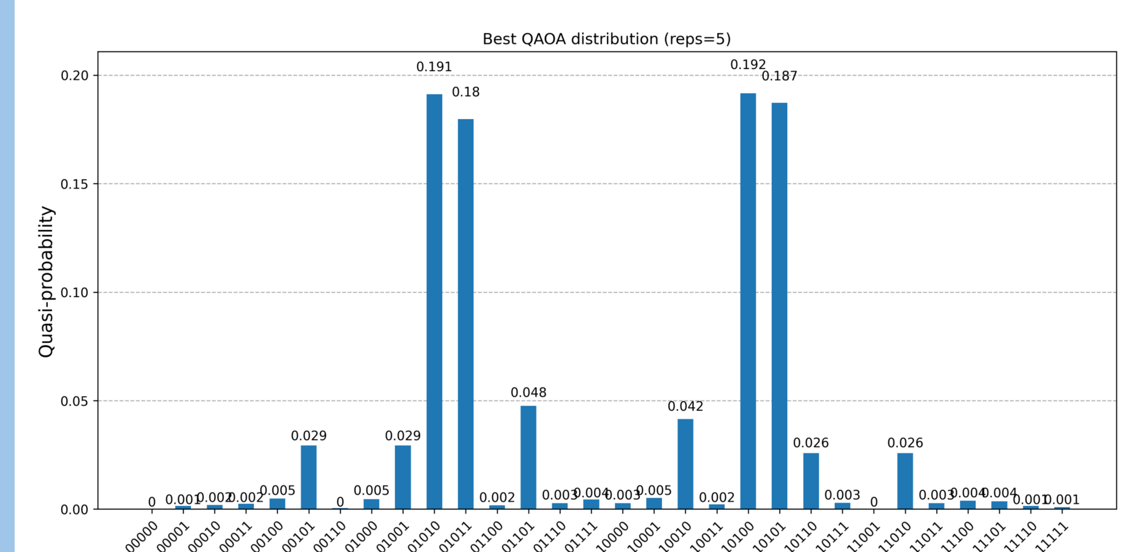
- Select edge with largest | M_{ij} |
- Correlate or anti-correlate variables
- Reduce graph by eliminating one variable
- Repeat until small instance (≤ 2 nodes)
- Solve final instance exactly

- **Key Insight**

RQAOA improves solution quality by decomposing the problem into smaller subproblems, at the cost of increased classical computation.

Results

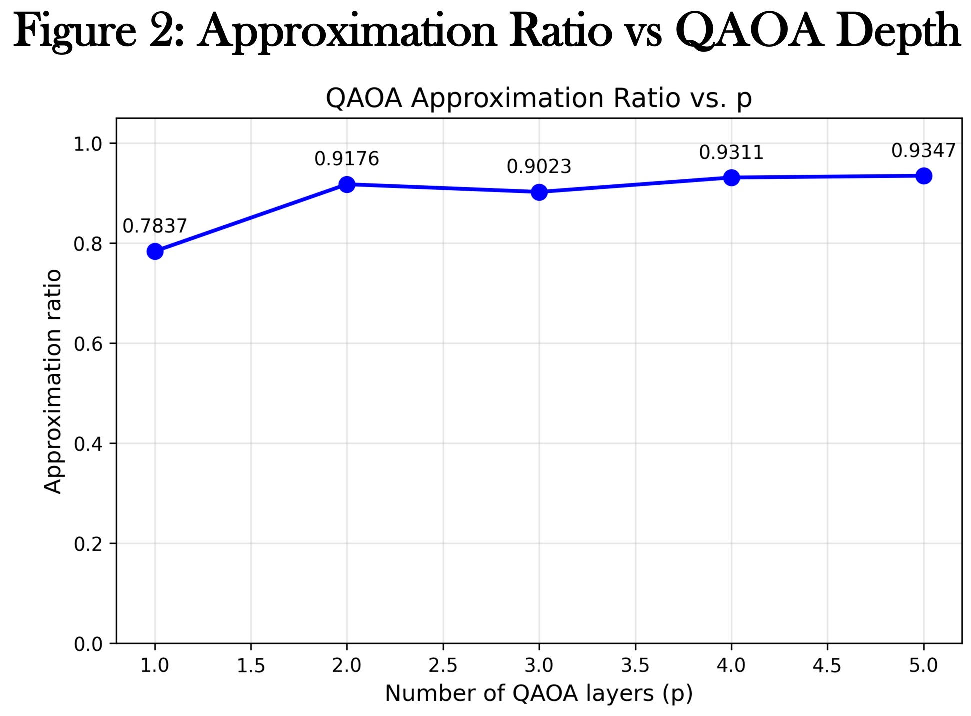
Figure 1: Probability Distribution of Bitstrings (p=5)



Histogram: bitstrings (0-31) vs probability. Highlight optimal cut(s).

Observation: The optimal solution receives significantly higher probability than suboptimal ones, indicating successful parameter learning.

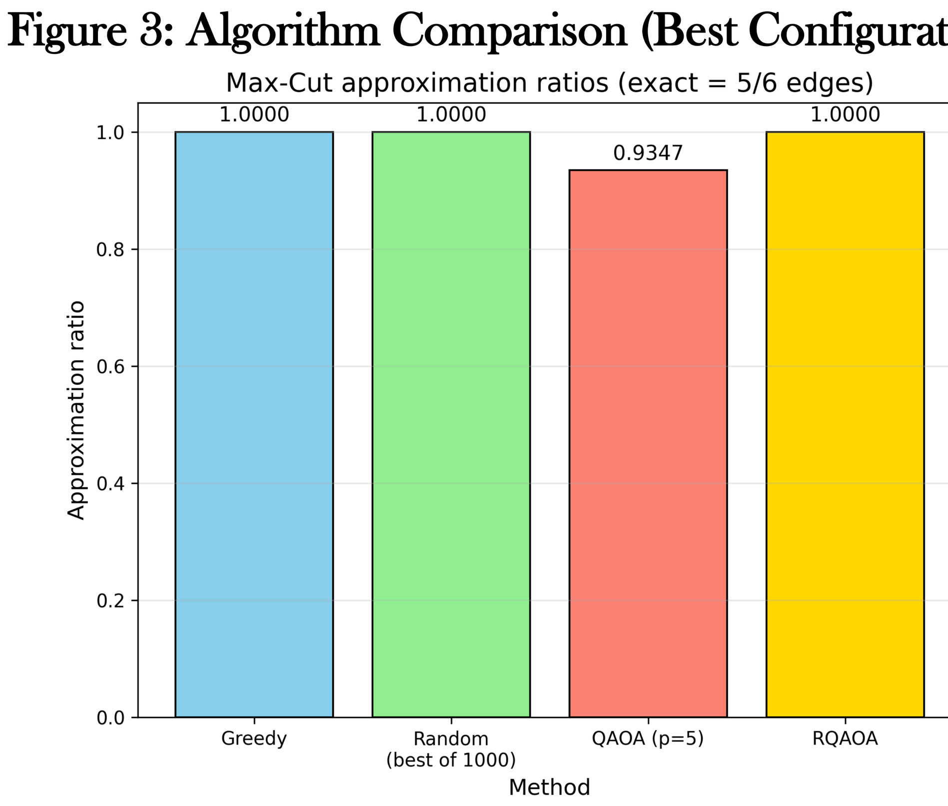
Figure 2: Approximation Ratio vs QAOA Depth $\boldsymbol{p}$



Approximation ratio vs QAOA depth p. Saturation observed at p = 4.

Observation: Approximation Ratio increases with depth, saturating around $p = 4$. Depth-5 yields marginal gain.

Figure 3: Algorithm Comparison (Best Configuration)



Observation: RQAOA achieves the exact optimal cut (AR = 1.000), outperforming standard QAOA (AR = 0.9347) and matching classical heuristics.

Table 1: Summary of Performance

Max-Cut Performance: QAOA vs RQAOA (Exact max cut = 5/6 edges)

Method	Approx. Ratio	Depth (p/reps)	Circuit Evaluations	Time (s)
QAOA (best p)	0.9347	5	100,000	38.2260
RQAOA	1.0000	1	22,500	44.7800

References

1. E. Farhi, J. Goldstone, and S. Gutmann, "A quantum approximate optimization algorithm," arXyh biv preprint, vol. arXiv:1411.4028, 2014. [quant-ph].
2. E. Bae and S. Lee, "Recursive QAOA outperforms the original QAOA for the MAX-CUT problem on complete graphs," arXiv preprint, 2023.